

Syllabus for

Academic year 2021/2022 >

ESS116 - Electrical circuits and systems

Elektriska nät och system

Syllabus adopted 2021-02-15 by Head of Programme (or corresponding)

Owner: [TKTFY](#)

7,5 Credits

Grading: TH - Pass with distinction (5), Pass with credit (4), Pass (3), Fail

Education cycle: First-cycle

Main field of study: Electrical Engineering, Engineering Physics

Department: 59 - MICROTECHNOLOGY AND NANOSCIENCE

Teaching language: Swedish

Application code: 57135

Open for exchange students: No

Maximum participants: 130

Only students with the course round in the programme plan

Credit distribution

Module	Sp1	Sp2	Sp3	Sp4	Summer course	No Sp	Examination dates
0111 Examination 7,5 c Grading: TH	1,5 c	6,0 c					Mon 10/01-2022 pm J, Wed 13/04-2022 am J, Fri 19/08-2022 pm J

In programs

[TKTFY ENGINEERING PHYSICS, Year 2 \(compulsory\)](#)

Examiner:

[Jan Grahm](#)

[Go to course homepage](#)

Eligibility

General entry requirements for bachelor's level (first cycle)

Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Specific entry requirements

The same as for the programme that owns the course.

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Course specific prerequisites

Linear algebra, real analysis, complex mathematical analysis, mechanics 1

Aim

The aim of the course is to provide the basic methodology in analysing analogue electrical and electronic circuits. This includes knowledge on the traditional passive circuit elements, sources, diodes, transistors and operational amplifiers. In addition, the course provides knowledge in how the electrical circuit operates as a system where concepts such as transfer function, frequency response and feedback are treated.

Learning outcomes (after completion of the course the student should be able to)

- Analyze passive linear electrical DC and stationary AC circuits
- Calculate transient signals in electrical circuits
- Analyze active electrical circuits based on diodes, transistors and opamps
- Gain insight into the electrical circuit as a system and be able to analyze time- and frequency response
- Understand concepts of feedback and stability
- Understand the design of an oscillator and an active filter
- Use computer-based tool for analyzing electrical circuits

Content

- General on electrical circuits. Definition of voltage and current source
- Passive elements: Resistance, capacitance, and inductance
- Power
- Kirchoff's laws
- Calculation methods for DC circuits
- Calculation methods for stationary AC circuits
- Impedance
- Complex power
- Calculation of transients in circuits with or without beginning energy.
- Introduction to linear time invariant systems. Transfer function.
- Diode circuits
- Transistor circuits (Bipolar Junction Transistor and field-effect transistor (MOSFET)). Quiescent point and small-signal circuits.
- Operational amplifier
- Analysis of circuits in time-and frequency domain
- Step- and pulse response
- Bode plot
- Feedback amplifiers. Stability (amplitude- and phase margin)
- Oscillator circuits
- Active filters

Organisation

- Lectures and exercises
- Laborations (3)
- "Dugga" (optional)

Literature

Elektriska nät. C. Davidson & P.G. Hofvenschiöld. Studentlitteratur 2001.
Analog elektronik. 3e uppl. B. Molin. Studentlitteratur 2020.

Examination including compulsory elements

Written exam and passed laborations. The optional "dugga" can give bonus credits for the written exam and the two extra exams to follow.

The course examiner may assess individual students in other ways than what is stated above if there are special reasons for doing so, for example if a student has a decision from Chalmers on educational support due to disability.